

| Question |     |                      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks Available |     |     |       |       |      |
|----------|-----|----------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |                      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 1.       | (a) | (i)<br>(ii)<br>(iii) |  | Number of cake cases (1)<br>Time of fall (1)<br>Speed up (1)                                                                                                                                                                                                                                                                                                                                                                                                                                | 1<br>1          | 1   |     | 3     |       | 3    |
|          | (b) | (i)                  |  | 0.75 [s] circled                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1               |     |     | 1     |       | 1    |
|          |     | (ii)                 |  | Sum = 0.55 + 0.56 + 0.55 + 0.58 = 2.24 (1)<br>Mean = $\frac{2.24}{4}$ = 0.56 [s] (1) Accept 0.6 s where this calculation is shown<br><br><b>Award only 1 mark</b> for the mean if 0.75 is included (mean = 0.60 [s])<br><b>Award only 1 mark</b> if no workings are shown and answer of 0.6 [s] or 0.598 [s]<br>Don't accept 0.59 [s]<br><br><b>Alternative:</b><br>$\frac{1.50}{2.7}$ (1)<br>= 0.56 [s] (1) accept 0.555 [s] or 0.6 [s]<br>Award 1 mark for $\frac{150}{2.7}$ = 55.5 or 56 | 1               | 1   |     | 2     | 2     | 2    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                              | Marks Available |          |          |           |          |           |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
|          | (c) | (i)  |  | All points plotted correctly < 1 small square tolerance award 2 marks (ignore (0,0))<br>4 points plotted correctly < 1 small square tolerance award 1 mark<br>3 or fewer points plotted correctly < 1 small square tolerance award 0 marks<br>Smooth curve passing through the points between 1-5 on the number of cake cases < 1 small square tolerance (1) |                 | 3        |          | 3         | 3        | 3         |
|          |     | (ii) |  | Increases (1)<br>Decreasing (1)                                                                                                                                                                                                                                                                                                                              |                 | 2        |          | 2         |          | 2         |
|          |     |      |  | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                      | <b>4</b>        | <b>7</b> | <b>0</b> | <b>11</b> | <b>5</b> | <b>11</b> |